WebClient Computing
jQuery Event

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Announcements

- Project-2
  - Developing a web page using Javascript
  - Recommend to develop from your project-1 output
  - HTML/CSS part is not going to be evaluated unless you make significant improvement
  - What to submit
    - Project output source code (HTML, CSS, Javascript file, …)
    - Feature description – The outcome is made from Javascript that contains lots of dynamic features, and I might miss some if you do not mention them
    - If you want to be reevaluated the HTML/CSS part (project-1), describe how you improved it

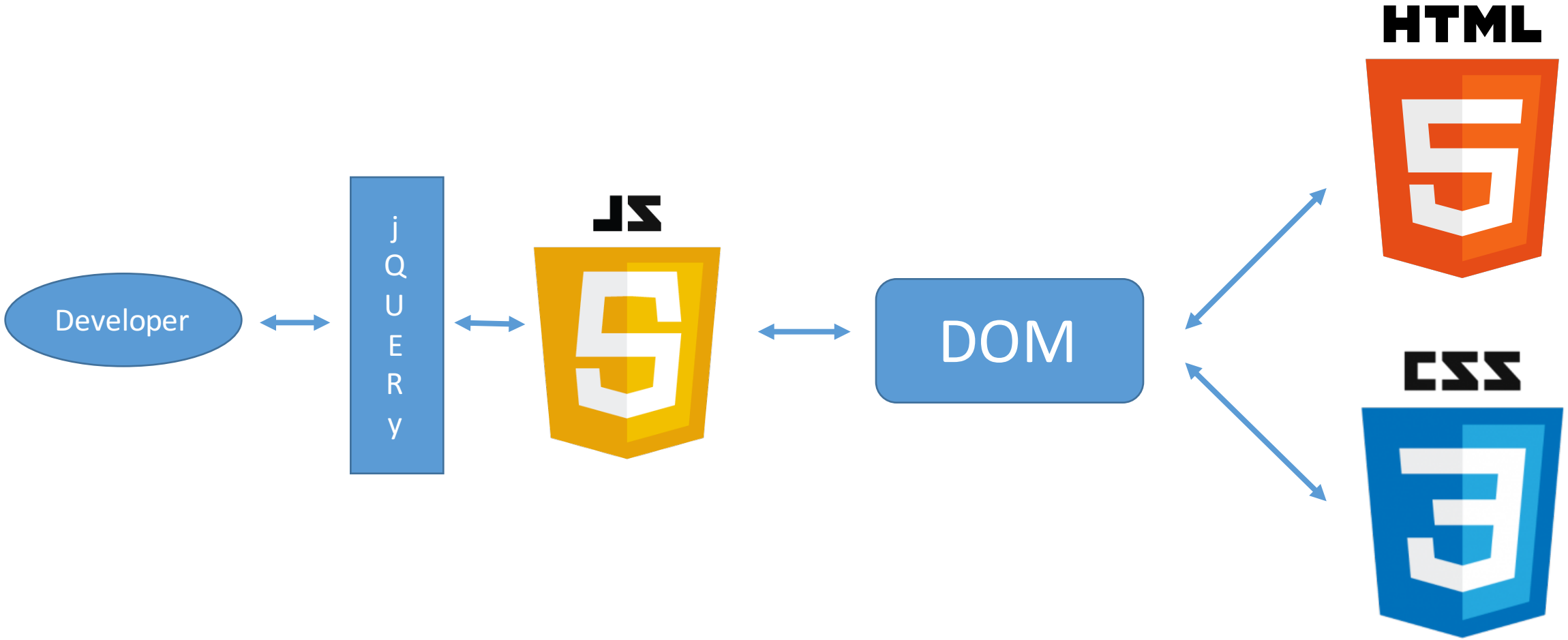
Announcements

- Project-2 evaluation is composed of
  - Code evaluation by the instructor
  - Interview
- Project-2 evaluation
  - How much you make the page dynamic
  - Using various Javascript, jQuery library
  - Understand how Javascript, jQuery works
  - The organization of the page
- Final exam schedule
  - June 16th or 17th
- Project-2 schedule
  - Tentative interview at June 22nd
  - I need about two days before the interview day for evaluation – June 20th

Announcements

- Homework-4 is out
- Course advertisement
  - Open source software in the next semester
    - two classes both from me
    - English course, but no worries
  - Java programming session in the summer break
    - Recommends to take if you have difficulties in using Java
    - The curse is provided on behalf of students (comments from conversation with faculties in CS)
  - I will teach object oriented programming (객체지향프로그래밍) for freshman – 5 classes are offered and I will take one class
    - You should be OK to take it if you want

jQuery, Javascript, and DOM



https://www.javacodegeeks.com/2016/01/7-ways-es2015-can-improve-your-javascript-programming.html
https://en.wikipedia.org/wiki/HTML

Traversing DOM Tree Using jQuery

- jQuery provides method to traverse DOM tree based on a current element
  - Child elements iteration
  - Going to parent node
  - Traversing sibling nodes

Traversing Siblings

- prev(optionalSelector), next(optionalSelector)
  - Retrieve the previous element on the same level (sibling)
- prevAll(optionalSelector), nextAll(optionalSelector)
  - Retrieve all the previous elements
- prevUntil(selector), nextUntil(selector)
  - Retrieve all the previous elements not including who meets selector

Traversing Siblings Example

- jquery_siblings.js jquery_sibling.html

```
</head>
<body>
  <div id = "container">
    <div style="color:blue" class="stop">First</div>
    <div style="color:blue">Second</div>
    <div style="color:blue">Third</div>
    <div id="target" style="color:blue">Fourth(Target ID)</div>
    <div style="color:blue">Fifth</div>
    <div style="color:blue">Sixth</div>
    <div style="color:blue" class="stop">Seventh</div>
  </div>
</body>
```

```
$('#target').prev().attr('style', 'color:red');
$('#target').prevAll().attr('style', 'color:red');
$('#target').prevUntil('.stop').attr('style', 'color:red');

$('#target').next().attr('style', 'color:green');
$('#target').nextAll().attr('style', 'color:green');
$('#target').nextUntil('.stop').attr('style', 'color:green');
```

Identifying Elements by Position

- index(element)
  - Returns an index of an element
- get(index)
  - Returns a DOM element with the given index
  - DOM element can be converted to jQuery object by $(DOM_ELEMENT)

jQuery Index Example

- jquery_index.js jquery_sibling.html

```html
<body>
  <div id = "container">
    <div style="color:blue" class="stop">First</div>
    <div style="color:blue">Second</div>
    <div style="color:blue">Third</div>
    <div id="target" style="color:blue">Fourth(Target ID)</div>
    <div style="color:blue">Fifth</div>
    <div style="color:blue">Sixth</div>
    <div style="color:blue" class="stop">Seventh</div>
  </div>
  <div id = "display"></div>
```

```javascript
var target_elem = $('#target');
var container_elem = $('#container');
var index = container.children().index(target_elem);

$('#display').text('Index is ' + index);

// get returns a DOM object (not a jquery object)
var found_element = container.children().get(index);

// convert DOM object to jqeury object
var jq_elem = $(found_element);
jq_elem.attr('style', 'color:red');
~
```

Handling Event with jQuery

- jQuery event handling
  - Remove event from HTML
  - Easy to update
  - Can be dynamic
- Access the event sender
  - Use this keyword
  - this object is DOM element (not jQuery object)
- $().
  - click
  - keypressed
  - mouseout

Event Handling With jQuery

- jQuery provides methods for registering event handlers
- Event handler registration using CSS selector, class, tag names, …
- Type of Events
  - click() – mouse click event
  - dblclick() – mouse double click event
  - mouseenter() – when a mouse pointer enters an element
  - mouseleave() – when a mouse pointer leaves an element
  - keypress() – when a keyboard is pressed
  - Full list - http://api.jquery.com/category/events/

Recap – Event Object

- Every event incudes an event object
  - EX: pageX and pageY for muse coordinates
  - EX: keyCode for a button pressed that triggered the event
  - Access the object using a keyword event

```
<!DOCTYPE html>
<html>
<head>
  <title>Javascript event</title>
  <script type="text/javascript" src="event_keyboard.js"></script>
</head>
<body>
<input type="text" size="40" onkeypress="key_pressed(event)">
<p id="key-demo"></p>
</body>
</html>
```

Handling Click Event With jQuery

- $(SELECTOR).click(function(){});
- When a mouse is clicked to the given selector, the given event handler is performed

```
$('#target-id').click(function (event) {
  output = "User clicked on " + event.pageX + "/" + event.pageY;
  $('#display').text(output);
})

$('.target-class').click(function() {
  output = "Class is clicked";
  $('#display').text(output);
})

$('div').click(function() {
      $('#display').text("div tag is clicked");
})
```

```
<body>
  <div id="target-id">ID - Click me</div>
  <div class="target-class">Class - Click me<div>
  <p id="display"></p>
  <script type="text/javascript" src="jquery_event.js"></script>
</body>
```

jquery_mouse_event.html/js

Compare Click Event Javascript - jQuery

- Javascript

```html
<!DOCTYPE html>
<html>
<head>
  <title>Javascript event</title>
  <script type="text/javascript" src="event.js"></script>
</head>
<body>

<p>Click the button to display the date.</p>

<button onclick="displayDate()">The time is?</button>

<p id="demo"></p>

</body>
</html>
```

```javascript
function displayDate() {
    document.getElementById("demo").innerHTML = Date();
}
```

jQuery

```javascript
$('#target-id').click(function (event) {
  output = "User clicked on " + event.pageX + "/" + event.pageY;
  $('#display').text(output);
})

$('.target-class').click(function() {
  output = "Class is clicked";
  $('#display').text(output);
})

$('div').click(function() {
      $('#display').text("div tag is clicked");
})
```

```html
<body>
  <div id="target-id">ID - Click me</div>
  <div class="target-class">Class - Click me<div>
  <p id="display"></p>
  <script type="text/javascript" src="jquery_event.js"></script>
</body>
```

Reference

- edx jQuery course
- w3schools.com
- http://api.jquery.com/